LAXMINARAYAN MOHANTA

Game Development Enthusiast & C++ Engineer

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PROFILE

I am a second-year B.Sc. (Hons) Mathematics and Computing student at the Institute of Mathematics and Applications, Bhubaneswar, with a genuine interest in game development and C++ engineering. I enjoy building things from first principles — whether that means a file-based record system or a calendar application that calculates dates centuries into the past — and I like understanding not just how to make something work, but why it works well. Alongside my coursework, I am currently part of a team developing a fingerprint-based access system for our campus, which has given me a taste of what it means to build software collaboratively. My mathematics background has sharpened the way I think through problems, and I am looking to build on that foundation as I move deeper into C++ and, eventually, game engine development.

EDUCATION

B.Sc. (Hons) Mathematics and Computing — IMA, Bhubaneswar *2025 – 2029*

Currently in my second year, building a strong grounding in both mathematics and computer science.

Higher Secondary Education (Class XII) — Odisha Adarsha Vidyalaya, Titia, Gopabandhu Nagar

Completed my higher secondary schooling here.

Secondary Education (Class X) — PM SHRI Kendriya Vidyalaya, Mahuldiha, Rairangpur

Completed my secondary schooling here.

PROJECTS

Student Database Management System — C++ *2025 – 2026*

- Built a console-based student record system in C++, using a custom data structure to hold each student's roll number, name, and marks across five subjects.
- Chose a two-level storage approach — a text file for each student alongside a central index file — so records could be listed quickly without scanning the file system directly.
- Gave the program a complete set of operations: adding, viewing, editing, and deleting records, along with a conflict-resolution menu for handling duplicate roll numbers gracefully.
- Added validation so marks always fall within a sensible 0–100 range, and built in a safeguard that exits the program after too many invalid menu choices in a row.
- Rewrote an early recursive approach to retrying input as a simple loop instead, once I realised repeated recursion could eventually overflow the stack.

Tools & Stack: C++, fstream, vector, algorithm

Calendar Application — C *2025*

- Wrote a console-based calendar generator in C that can render any Gregorian year — including years before the Common Era — with correct leap-year handling throughout.
- Worked out the date-arithmetic logic myself, calculating the first day of a year and the day of the week for any date without relying on the standard <time.h> library.
- Put together a special-day lookup feature covering a wide range of Indian and international observances, holidays, and awareness days across all twelve months.
- Added the ability to attach a personal note to any date and read it back later, with each note saved as its own text file.

- Included date validation and a listener for the escape key, so the program always exits cleanly if the user wants out.

Tools & Stack: C, File I/O, conio.h

Fingerprint-Based Campus Access System — Team Project (Ongoing) *2026*

- Working with a small team to design and build a fingerprint-based system for tracking entry and exit across campus.
- Taking on my share of the design and implementation work as one of the core contributors to the project.

Status: In Progress

RELEVANT COURSEWORK

2nd Year

- Linear Algebra
- Abstract Algebra
- Design & Analysis of Algorithms (DAA)
- Real Analysis
- Probability & Statistics
- Ordinary Differential Equations

1st Year

- Operations Research
- Real Analysis
- Abstract Algebra
- Introduction to Python
- Programming and Data Structures
- Discrete Mathematics, Algebra and Number Theory

SKILLS

- Programming: comfortable working in both C and C++.
- Core concepts: file handling, data structures, object-oriented programming, and algorithm design.
- Soft skills: clear communication, adaptability, problem-solving, and working well within a team.
- Particularly drawn to game development, systems programming, and C++ engineering.

LANGUAGES

- English — Proficient
- Odia — Native
- Hindi — Proficient

HOBBIES & INTERESTS

- An enthusiastic gamer — I've spent a lot of time across titles like God of War, Clash of Clans, Free Fire, BGMI, Kingshot, Asphalt 9, and Sekiro.
- I enjoy building small C++ projects in my free time, mostly as a way to keep learning and try out new ideas.